

CHITRAL, Pakistan

WMO: 415060

Lat: 35.883N

Lon: 71.800E

Elev: 1500

StdP: 84.56

Time Zone: 5.00 (E05)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.7	-1.4	-13.1	1.4	8.6	-11.0	1.8	8.1	9.0	8.3	8.4	8.3	0.3	N/A	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	37.4	20.7	36.2	20.6	34.9	20.3	23.9	33.2	22.9	32.3	22.1	31.5	2.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	19.3	29.7	20.2	17.9	28.5	19.3	16.9	27.9	81.4	33.2	76.7	32.3	73.1	31.5	32.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.8	6.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-5.9	26.1	1.8	2.0	-7.2	27.5	-8.2	28.7	-9.2	29.8	-10.5	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	4.5	5.9	10.8	16.1	20.7	25.8	28.3	26.5	22.2	16.1	9.8	6.4
	DBStd	8.69	2.45	2.96	3.48	3.39	3.10	2.90	2.17	2.49	2.93	3.12	2.58	2.30
	HDD10.0	469	169	118	32	2	0	0	0	0	0	0	33	114
	HDD18.3	1818	428	347	233	82	12	1	0	1	4	86	255	371
	CDD10.0	2719	0	4	58	184	333	474	568	512	366	188	28	2
	CDD18.3	1026	0	0	0	14	87	224	310	254	120	16	0	0
	CDH23.3	13126	0	0	9	221	1081	2832	4138	3105	1456	280	4	0
	CDH26.7	6657	0	0	0	57	449	1507	2356	1613	593	83	0	0
Wind	WSAvg	1.5	1.5	1.2	1.3	1.2	1.2	1.7	2.2	2.0	1.7	0.9	0.9	1.6
Precipitation	PrecAvg	539	49	83	112	91	48	15	13	12	19	28	35	32
	PrecMax	757	144	226	258	160	126	46	92	44	110	240	193	119
	PrecMin	247	0	9	18	15	4	1	1	0	1	0	0	0
	PrecStd	105	40	54	60	36	33	13	17	11	22	46	39	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	19.0	24.5	31.1	34.7	39.0	39.4	37.9	34.4	31.9	23.8	17.3
		MCWB	5.1	7.8	12.9	16.7	19.0	20.5	21.3	21.5	21.7	16.6	11.9	7.4
	2%	DB	13.1	15.5	22.0	28.3	33.0	37.1	37.7	36.3	33.0	29.0	20.0	15.0
		MCWB	4.4	7.1	11.3	15.5	18.1	19.4	21.4	21.7	20.6	16.2	10.7	6.5
	5%	DB	11.6	13.8	20.0	25.8	31.4	35.5	36.4	34.9	31.7	26.5	18.2	13.4
		MCWB	4.2	6.4	10.3	14.1	17.6	18.8	21.0	21.4	19.8	15.2	10.0	5.5
	10%	DB	10.0	12.0	18.1	23.6	29.6	33.8	35.2	33.6	30.2	24.0	16.4	11.9
		MCWB	3.5	5.5	9.3	13.3	16.8	18.3	20.7	21.0	19.1	14.2	9.2	4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	9.6	13.8	17.8	21.7	22.6	26.2	25.1	24.8	20.1	13.7	9.0
		MCDB	12.2	14.2	23.1	28.6	32.4	34.0	34.4	33.8	33.1	27.4	21.4	14.7
	2%	WB	5.7	8.6	12.4	16.2	19.5	21.2	23.8	23.8	22.9	17.5	11.9	7.5
		MCDB	10.8	13.4	19.8	26.2	30.7	32.7	34.4	32.6	31.0	26.2	17.7	12.9
	5%	WB	4.8	7.5	11.2	15.1	18.2	20.2	22.7	22.9	21.0	15.9	10.9	6.4
		MCDB	9.3	12.1	17.9	23.5	29.0	31.6	33.3	31.4	29.4	24.7	16.3	11.8
	10%	WB	4.1	6.3	10.2	14.2	17.2	19.3	21.9	22.1	19.8	14.6	9.8	5.4
		MCDB	8.5	10.5	16.3	21.6	27.5	31.2	31.9	31.0	28.2	22.8	15.0	10.6
Mean Daily Temperature Range	5% DB	MDBR	9.7	9.3	11.0	12.8	14.7	15.2	13.7	13.6	15.0	15.1	12.6	10.8
		MCDBR	12.8	12.3	14.5	16.9	17.8	17.2	15.4	15.0	16.4	18.6	16.0	13.6
		MCWBR	6.2	5.7	6.0	6.8	7.0	5.4	4.5	4.2	5.9	7.1	7.9	6.7
	5% WB	MCDBR	10.8	10.2	13.4	15.0	16.4	15.5	13.9	13.6	15.5	17.1	13.5	12.4
		MCWBR	5.9	5.3	6.0	6.6	7.6	5.8	5.4	5.6	7.2	7.8	7.4	6.8
Clear-Sky Solar Irradiance	taub	0.346	0.389	0.452	0.475	0.476	0.481	0.489	0.461	0.404	0.394	0.385	0.346	
	taud	1.998	1.867	1.773	1.834	1.917	1.939	1.949	2.046	2.169	2.093	2.013	2.059	
	Ebn at Noon	815	821	805	813	817	811	801	816	847	812	764	788	
	Edh at Noon	133	172	207	206	192	188	185	165	138	135	131	118	
All-Sky Solar Radiation	RadAvg	2.58	3.32	4.65	5.78	6.58	7.25	7.10	6.56	5.66	4.29	2.92	2.40	
	RadStd	0.20	0.36	0.35	0.49	0.52	0.45	0.36	0.45	0.36	0.40	0.32	0.24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page